A Greedy Band Selection Strategy Based on Dual-Frequency Collaborative Feature Fusion for Unsupervised Hyperspectral Band Selection

Zexin Zeng, Cuiping Shi, *Member, IEEE*, Liguo Wang, *Member, IEEE*, and Haizhu Pan

***Abstract*—Band selection plays a crucial role as a preprocessing step in hyperspectral image (HSI) processing tasks. Currently, most band selection methods tend to fuse spatial spectral features, directly from the original spectral domain, without fully exploring the distinct information embedded in different frequency components of the spectrum. In this study, a greedy band selection (GBS) strategy based on dual-frequency collaborative feature fusion (GSDFC) is proposed for unsupervised band selection of HSIs. First, a dual-frequency collaborative feature fusion module (DFCFFM) is designed to extract and fuse the spatial spectral features of two frequency domain components separately. Second, to further enhance the effectiveness of DFCFFM and learn important features in low-frequency components adaptively, a grouping spatial attention module (GSAM) is introduced, which effectively captures global and local spatial dependencies. Finally, a greedy selection strategy is constructed by utilizing the mutual information (MI) of the fused features and the Pearson's correlation coefficient (PCC) of the original data to select the optimal subset of bands. The proposed GSDFC band selection method is validated by classification. Extensive experiments have shown that, compared with some advanced methods, the proposed GSDFC method can achieve the best classification performance on three public datasets: Indian Pines, Pavia University, and Houston 2013, which fully demonstrates the effectiveness of the proposed GSDFC method. The related code will be released at https://github.com/Zengzx716/GSDFC**

***Index Terms*—Dual-frequency collaborative, feature fusion, greedy selection strategy, hyperspectral images (HSIs), unsupervised band selection.**

I. Introduction

HYPERSPECTRAL images (HSIs) are collections of multiple narrowband images, each representing a radiation signal within a distinct wavelength range. Each band reflects the scene composed of various objects imaged at different wavelengths [1]. In recent years, HSIs have been widely applied in various fields such as environmental monitoring [2], fine-grained classification [3], and resource exploration [4]. The collection of HSIs involves recording multiple spectral bands within a specific wavelength range. For some applications such as land cover classification, HSIs with their rich spectral bands can accurately perceive and identify land cover [1]. However, as the dimensionality of data increases, the model becomes overly complex, leading to overfitting and the well-known "Hughes phenomenon," which is also known as the curse of dimensionality [5]. Therefore, dimensionality reduction has become an important step in HSI data processing.

Currently, dimensionality reduction techniques are mainly implemented using two methods: feature extraction [6] and feature selection, also known as band selection [7]. Traditional band selection methods include three main categories: filter methods [8], [9], wrapper methods [10], [11], [12], [13], and embedded methods [14], [15]. Meanwhile, traditional methods can also be further categorized into the following four types: ranking [16], [17], searching [18], [19], clustering [20], [21], [22], [23], and sparse representation methods [24], [25].With the development of technology, deep learning has made significant progress in various fields [26], such as graph neural networks [27], Transformers [28], and Mamba [29]. Most of them are all excellent in band selection methods [30], [31], [32].

According to whether labeled samples are used, existing band selection methods can also be categorized into supervised [33], semisupervised [34], [35], [36], and unsupervised methods. Compared with supervised and semisupervised methods, unsupervised band selection methods are applicable to more downstream tasks, which do not rely on prelabeled data or their independence from labeled data constraints. Unsupervised band selection methods do not rely on any known category information and can autonomously explore the intrinsic structure of sample data, thereby achieving efficient band selection, which is currently a hot topic in band selection research [37], [38], [39], [40], [41].

Currently, pixel classification is an important application for HSIs. Based on the integration of features, existing band selection techniques can be categorized into spatial-based,

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Zexin Zeng is with the Department of Communication Engineering, Qiqihar University, Qiqihar 161000, China (e-mail: 2022910311@qqhru.edu.cn).

Cuiping Shi is with the College of Communication and Electronic Engineering, Qiqihar University, Qiqihar 161000, China, and also with the College of Information Engineering, Huzhou University, Huzhou 313000, China (e-mail: shicuiping@zjhu.edu.cn).

Liguo Wang is with the College of Information and Communication Engineering, Dalian Nationalities University, Dalian 116000, China (e-mail: wangliguo@hrbeu.edu.cn).

Haizhu Pan is with the College of Computer and Control Engineering, Qiqihar University, Qiqihar 161000, China (e-mail: panhaizhu@qqhru.edu.cn).

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spectral-based, or a hybrid of spectral and spatial information. BSNet-Conv was proposed by Cai et al. [42] based on convolutional neural networks, utilizing 2-D convolutional layers and global pooling layers to simultaneously extract spectral and spatial information. An algorithm for spectral clustering named efficient graph convolutional self-representation (EGCSR) was proposed by Cai et al. [43], which utilize spectral graph embedding to capture structural information among spectral data. A band selection algorithm called region-wise latent feature fusion and graph filter embedded subspace clustering (RFGEC) was proposed by Feng et al. [44], which effectively capture spatial information by segmenting the original image into multiple homogeneous regions and fusing the features of these regions into a low-dimensional latent space. A new HSI band selection method named DARecNet-BS was proposed by Roy et al. [45]. Compared with traditional convolutional neural networks, DARecNet-BS introduces a pixel attention module and channel attention module, which can automatically learn and extract useful spectral and spatial features. A region-based subspace clustering method with low-dimensional feature representation and graph regularization for HSI band selection (GRSC) was proposed by Wang et al. [46], which not only explore the spectral correlations among all bands but also preserve the spatial structural information within different subregions. An unsupervised band selection method called dual global–local attention network (DGLAnet) was proposed [47], which fully uses the global and local features of spatial and spectral information. By leveraging the interactions between pixels and superpixels to embed spatial and structural information into the model, a novel hyperspectral band selection method based on global–local graph self-representation (tensorial global–local graph self-representation, TGSR) was proposed [48], which improves the accuracy of band selection. A method named joint spatial and spectral graph consistent self-representation (JSSGCSR) was proposed by Ma et al. [49], which effectively fused spatial and spectral information while leveraging graph convolutional networks to capture structured information in the data. Cai et al. [50] proposed an unsupervised HSI band selection method-based deep learning called spectral–spatial cross-dimensional attention network for hyperspectral band selection (SSANet-BS).

The existing methods for selecting spatial spectral bands suffer from key issues such as loss of details, sensitivity to noise, and high redundancy due to neglecting frequency-domain features. Specifically, first, when relying solely on the original spatial spectral features, edge, and texture details in high-frequency components are easily ignored, leading to blurred segmentation boundaries and increased the probability of misclassification in classification tasks. Second, the direct involvement of unseparated high-frequency noise in band selection can introduce large errors. Finally, most existing methods have not fully considered the continuity characteristics of low-frequency spectrum, and the spectral redundancy between adjacent bands is still high, further reducing the discriminability of band subsets.

From the above analysis, existing spatial–spectral band selection methods often focus on spatial or spectral information, while ignoring the critical role of frequency-domain features. For instance, some methods such as [44] relying solely on spatial attention mechanisms fail to capture long-range dependencies and detailed frequency components. Although some approaches such as [45], [46], [47], [48], [49], [50] attempted to integrate spatial and spectral relationships, they often ignored the high-frequency details and structures embedded in HSIs. To overcome these limitations, this article proposes a greedy strategy based on dual-frequency collaborative feature fusion (GSDFC) for HSI band selection, which leverages wavelet transform to extract frequency domain features, aiming to capture more local and global features in HSIs by learning frequency-domain features and utilizes the greedy band selection (GBS) strategies to select the optimal band subsets. Considering that the traditional spatial attention module (SAM) cannot effectively capture long-range or cross-region dependencies when capturing local contextual information, in this article, a grouping SAM (GSAM) is introduced to effectively capture global and local spatial dependencies. To overcome the limitations of a single matrix, a greedy selection strategy is proposed that effectively combines the Pearson's correlation coefficient (PCC) and mutual information (MI) as two metrics, thus achieving better selection results. To summarize, the main contributions of this work are as follows.

1) A dual-frequency collaborative feature fusion module (DFCFFM) is constructed, which dynamically balances spatial details and spectral continuity, effectively achieves the features fusion of spatial and spectral information in HSIs by learning deep features and detailed features with different frequency components.
2) A greedy selection strategy has been proposed, which effectively integrates different statistical measurement methods while capturing significant linear correlations and complex nonlinear relationships between bands, making the selected bands both informative and relatively independent.
3) A GBS strategy based on dual-frequency collaborative feature fusion (GSDFC) is proposed for unsupervised band selection of HSIs, which achieve unsupervised band selection collaborative optimization and can effectively improve the accuracy of band selection.

The remainder of this article is organized as follows. Section II reviews related work, Section III presents the proposed method, Section IV demonstrates the experimental results and analysis, and Section V gives the concludes.

II. Related Work

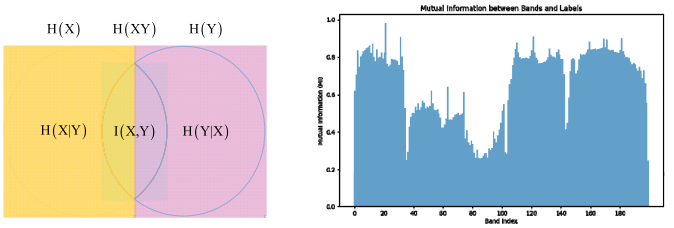
In this section, we briefly introduce some of the principles involved in the proposed method, including MI and discrete wavelet transform (DWT), to provide a comprehensive understanding of subsequent work.

A. Mutual Information

Assuming the input data is denoted as X and the corresponding label is denoted as Y, the MI between the data and label is defined as

$$I(X,Y)=\sum_{y=1}^{i}\sum_{x=1}^{j}p(x,y)\log\frac{p(x,y)}{p(x)\,p(y)} \tag{1}$$

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(a) Venn diagram of MI

(b) MI between bands and labels of Indian Pines

Fig. 1. MI diagram. (a) Venn diagram of MI. (b) MI between bands and labels of Indian Pines.

where x, y are the pixel values, i, j are the maximum pixel values of the input data and label, respectively. $p(x, y)$ is the joint probability distribution function, $p(x)$ and $p(y)$ are the marginal probability distribution functions. As a method for measuring the degree of mutual dependence between two variables, MI can quantify the linear or nonlinear correlation between two variables. When the two variables are completely independent, $I(X, Y) = 0$. Because of its ability to capture nonlinear relationships, MI has been widely used in feature selection, and features with higher MI values for classification targets are more suitable for classification [34].

The MI can also be calculated using entropy. As shown in the Venn diagram in Fig. 1(a), the left circle represents the entropy of the data, the right circle represents the entropy of the label, and the intersection represents the joint entropy of both. Therefore, the MI can be calculated via $I(X, Y) = H(X) + H(Y) - H(XY)$.

B. Minimum Redundancy and Maximum Relevance

Minimum redundancy and maximum relevance (mRMR) is a filter-based feature selection framework aimed at selecting feature subsets from a high-dimensional feature space that are highly correlated with the target variable and have the minimum redundancy [51]. mRMR decomposes the feature importance score into two complementary parts, namely, maximum relevance, and minimum redundancy. By jointly optimizing relevance and redundancy, key information is retained while improving model efficiency.

Assuming the candidate features is denoted as X_i, the target varience is denoted as Y and the selected feature subset is denoted as S, the mRMR score is defined as

$$f^{\mathrm{mRMR}}(X_i) \leftarrow F(Y, X_i) - \frac{1}{|S|} \sum_{X_s \in S} F(X_s, X_i) \tag{2}$$

where $F(\cdot)$ represents the calculation function of maximum relevance and $|S|$ is the size of selected feature subset.

C. Discrete Wavelet Transform

DWT can decompose data into different frequency interval components and is widely used in image processing [52]. In deep learning, max pooling is often used for down-sampling, but it can easily destroy the basic object structure [53]. DWT decomposes the original image into low-frequency components and high-frequency components, where the low-frequency components represent the basic object structure and the high-frequency components contain a large amount of image details. Compared with max pooling, DWT can preserve high-frequency details during down-sampling, better capturing subtle changes and features of data.

Two-dimensional DWT and 3-D DWT can both be implemented through a series of 1-D DWT along different dimensions of data. For 2-D signals X_{2d}, 1-D DWT is performed on their rows and columns, that is,

$$\begin{aligned} \mathrm{X}_{2d}^{ll} &\leftarrow L\mathrm{X}_{2d}L^T, \quad \mathrm{X}_{2d}^{lh} \leftarrow H\mathrm{X}_{2d}L^T \\ \mathrm{X}_{2d}^{hl} &\leftarrow L\mathrm{X}_{2d}H^T, \quad \mathrm{X}_{2d}^{hh} \leftarrow H\mathrm{X}_{2d}H^T \end{aligned} \tag{3}$$

where H and L represent the high-pass and low-pass filters, respectively. Therefore, a low-frequency component X^{ll} and three high-frequency components $\mathrm{X}^{lh}, \mathrm{X}^{hl}$, and X^{hh} can be obtained through the 2-D DWT. Similarly, using the 3-D DWT for 3-D signals can result in one low-frequency component X^{lll} and seven corresponding high-frequency components.

III. Methodology

Most existing methods focus only on spatial or spectral features, making it difficult to distinguish between noise and effective signals, and unable to balance details and structural information. Frequency-domain features are the key bridge connecting spatial structure and frequency information. Therefore, through multiscale decomposition using wavelet transform, high-frequency components capture high-frequency details of the image, while low-frequency components represent the global low-frequency structure, optimizing band selection from two dimensions. Fig. 2 illustrates the overall architecture of the proposed method. The proposed method comprises two main parts: a DFCFFM and GBS. The DFCFFM employs DWT to sample data into different frequency intervals, obtaining high-frequency and low-frequency components of the original data. The obtained components are then used to learn deep and detailed features through different convolutions. Finally, the learned features are fused. GBS constructs a greedy strategy to effectively combine the PCC of the original data and MI of the fused features to select the optimal subset of bands.

For the original HSI data $\mathrm{X} \in \mathbb{R}^{h \times w \times b}$, where h and w represent the height and width of the original data, respectively, b denotes the number of original bands. After padding the boundary pixels, extract patches around each central pixel to obtain an HSI cube $\mathrm{X}_p \in \mathbb{R}^{p \times p \times b}$, where $p \times p$ represents the spatial size; then, X_p is used as the input data for down-sampling.

A. Dual-Frequency Collaborative Feature Fusion Module

The proposed DFCFFM aims to fully utilize the deep features of low-frequency component and the detailed features of high-frequency components, further achieving the fusion of spatial–spectral information features in HSI data. As shown in Fig. 2, DFCFFM mainly consists of DWT downsampling, spatial–spectral feature extraction, and feature fusion.

For the input HSI cube X_p, 2-D, and 3-D DWT are adopted for downsampling. After 2-D DWT, a low-frequency component X_p^{ll} containing the main structural information and

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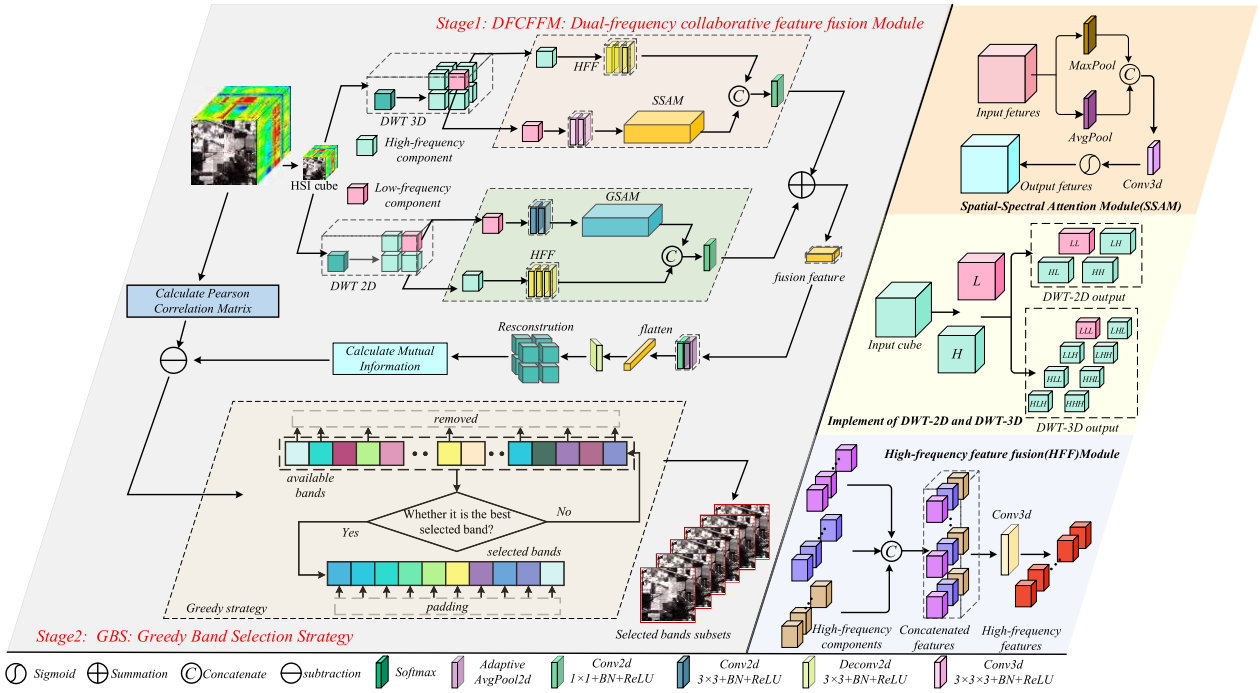


Fig. 2. Overall architecture of the proposed method, including two main stages: DFCFFM and GBS.

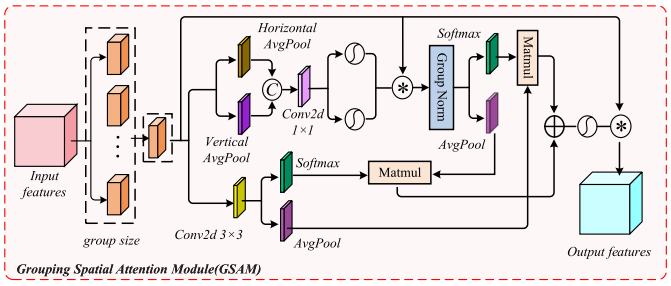


Fig. 3. Structure of GSAM.

three high-frequency components X_p^{lh}, X_p^{hl}, and X_p^{hh} containing detailed features are obtained. The other branch produces eight components after 3-D DWT sampling, namely one low-frequency component X_p^{lll} and seven high-frequency components $X_p^{llh}, X_p^{lhl}, X_p^{lhh}, X_p^{hll}, X_p^{hhl}, X_p^{hlh}$, and X_p^{hhh}. By leveraging the powerful local feature learning capability of convolution kernels, the low-frequency components obtained from the two branches are fed into two-layer 2-D CNNs and 3-D CNNs, respectively, for feature extraction, resulting in two output features F^{ll} and F^{lll}.

In order to adaptively learn important features in low-frequency components and enhance the effectiveness of feature extraction, a GSAM is introduced, which effectively captures global and local spatial dependencies through grouping and multiscale parallel subnetwork structures without channel dimensionality reduction. As shown in Fig. 3, for the input features, GSAM divides them into g subfeatures $F_g^{ll} \in \mathbb{R}^{(p/2)\times(p/2)\times(b//g)}$ in the channel dimension, where $g \ll b$. The grouped features adopt three parallel branches: two parallel 1×1 branches and one 3×3 branch. The two parallel 1×1 branches first employ 1-D global pooling (along the horizontal and vertical directions, respectively) to extract global information, then pass through 1×1 convolutional layers for feature fusion, and finally generate attention maps using the Sigmoid function. The other 3×3 branch employs a 3×3 convolution kernel to capture local spatial information, followed by a Softmax normalization. The attention maps generated by the first two branches are combined through matrix multiplication to produce the final feature representation. By using cross spatial learning methods to fuse the obtained feature maps, a global attention map is generated to capture pixelwise pairing relationships. The final attention map is combined with the original input feature map through elementwise summation and multiplication operations to generate the final output feature F^{ll}.

It is worth noting that for 3-D low-frequency components, the spectral SAM (SSAM) can operate simultaneously in both spectral and spatial dimensions, comprehensively capturing interband and pixelwise spatial correlations and reducing redundant features. The output features of SSAM can be obtained through the following operation, where $f^{a\times a\times a}$ denotes a convolution operation with a kernel size of $a \times a \times a$:

$$\begin{aligned} M^{ll} &\leftarrow \sigma\left(f^{a\times a\times a}\left(\left[\text{MaxPool}\left(F^{lll}\right); \text{AvgPool}\left(F^{lll}\right)\right]\right)\right) \\ f^{lll} &\leftarrow M^{ll} \odot F^{lll}. \end{aligned} \tag{4}$$

All high-frequency components obtained by DWT are processed using a high-frequency feature fusion module (HFF). Specifically, first concatenate all high-frequency components along the channel dimension, and then perform 1×1 convolution, high-frequency feature components can be obtained by

$$F^{hh} \leftarrow \text{Conv}_{1\times 1}\left(\text{Concat}\left(X^{lh}, X^{hl}, X^{hh}\right)\right)$$

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$$\mathrm{F}^{hhh} \leftarrow \mathrm{Conv}_{1\times1\times1}\left(\mathrm{Concat}\left(\mathrm{X}^{llh}, \mathrm{X}^{lhl}, \mathrm{X}^{lhh}\right.\right. \left.\left.\mathrm{X}^{hlh}, \mathrm{X}^{hhl}, \mathrm{X}^{hll}, \mathrm{X}^{hhh}\right)\right). \quad (5)$$

Following, the obtained high-frequency features are concatenated with the low-frequency features along the channel dimension. Subsequently, after two 1×1 convolution processes, the extracted features are mapped to a unified channel dimension. Finally, the two features, F_{2d} and F_{3d}, are fused, and the fused feature can be represented as

$$\mathrm{F}_{\mathrm{fusion}} \leftarrow \omega \cdot (\mathrm{F}_{2d} \oplus \mathrm{F}_{3d}) \quad (6)$$

where $\oplus$ denotes the elementwise summation, and ω is a manually set weight coefficient. The pseudocode of the implementation process of DFCFFM is given in Algorithm 1.

Algorithm 1 Implemental Process of DFCFFM

Input : Hyperspectral image data $\mathbf{X} \in \mathbb{R}^{h\times w\times b}$, patch data $\mathbf{X}_p \in \mathbb{R}^{p\times p\times b}$, wavename = 'db2', group = 8, coefficient hsi = 0.5

Output: Reconstructed hyperspectral image $\mathbf{X}'$

Initialize:
    3D HSI encoder HSI_Encoder_3D
    2D HSI encoder HSI_Encoder_2D
    Reconstruction network RectNet $(64, l_1 = X.shape[2])$
    Step 1: 3D Spectral–spatial Feature Extraction
$x_hsi_3d \leftarrow$HSI Encoder 3D$(\mathbf{X}_p)$
    // Apply 3D discrete wavelet transform
    // Process low-frequency components with 3D CNN
    // Apply spatial–spectral attention
    // Process high-frequency components
    //Fuse and transform features to 2D
Step 2:2D Spatial Feature Extraction
$x_hsi_2d \leftarrow$HSI Encoder 2D$(\mathbf{X}_p)$
    // Apply 2D discrete wavelet transform
    // Process low-frequency components with 2D CNN
    // Apply GSAM
    // Process high-frequency components
    //Fuse multiscale features
Step 3: Multibranch Feature Fusion
$x_hsi \leftarrow x_hsi_3d + x_hsi_2d$ 21: $x_cnn \leftarrow$ weight$_hsi \times x_hsi$ 22: Step 4: Feature Transformation
$x \leftarrow$ flatten $(x_cnn, \mathrm{start}_dim = 2)$
$x \leftarrow$ seq2img (x)
Step 5: Image Reconstruction
$\mathbf{X}_\mathbf{rec} \leftarrow$ RectNet (x)
    // Unsample features to original spatial size
    // Generate final reconstructed HSI
Return $\mathbf{X}_\mathbf{rec}$;

B. Greedy Band Selection

Greedy strategy is a method of making local optimal choices at each step, with the expectation that these local optimal choices will lead to a global optimal solution. The proposed GBS method aims to select the most representative bands from HSI data based on the mRMR framework. By combining the PCC and the MI between bands and labels, the optimal band combination is selected using a greedy strategy.

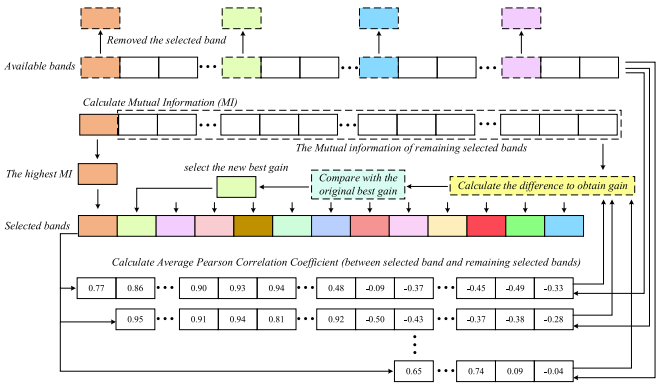


Fig. 4. Process of the proposed GBS.

Reshape the original input data $\mathrm{X} \in \mathbb{R}^{h\times w\times b}$ to obtain a new input $\mathrm{X}_{\mathrm{reshape}} \in \mathbb{R}^{hw\times b}$, and calculate the PCC between all bands, where $\mathrm{pcc}_{i,j}$ represents the PCC between the ith band and the jth band, that is,

$$\mathrm{pcc}_{i,j} \leftarrow \frac{\sum_{s=1}^{n}\left(x_{s,i}-\bar{x}_i\right)\left(x_{s,j}-\bar{x}_j\right)}{\sqrt{\sum_{s=1}^{n}\left(x_{s,i}-\bar{x}_i\right)^2 \sum_{s=1}^{n}\left(x_{s,j}-\bar{x}_j\right)^2}} \quad (7)$$

where $x_{s,i}$ represents the value of the sth sample in the ith band and $\bar{x}_i$ represents the mean value of the ith band. In addition, the fused features obtained through DFCFFM are flattened to obtain a 2-D array, where each sample corresponds to all bands of a cube. Following, the MI between each band and label is calculated.

The process of the proposed GBS method is illustrated in Fig. 4. Initialize two sets: one is the empty set selected_bands used to store the selected band indices, and the other is the set available_bands containing all possible selected band indices. Initially, when selected_bands is empty, select the band index with the highest MI from the available_bands and fill it to the selected_bands. The selected band index is then removed from the available_bands. When a band has already been selected, calculate the average correlation coefficient between that band and all selected bands, and subtract this average correlation coefficient from the MI of the bands as the gain, that is,

$$\mathrm{gain}_b \leftarrow \mathrm{MI}(b) - \mathrm{avg}\left(\mathrm{pcc}_{i,b}\right). \quad (8)$$

If the calculated gain for the current band gain_b exceeds the previously established optimal gain threshold, then update the optimal gain value, via

$$\mathrm{gain}_{\mathrm{best}} \leftarrow \mathrm{gain}_b. \quad (9)$$

At the same time, append band b, which yields the optimal gain, to the selected_bands. Through iterative updates, the current loop is terminated when the number of selected bands is greater than or equal to the preset number of selected bands k. Simultaneously, the optimal band subset is added to the selected_bands and removed from the available_bands. Finally, a list of selected band subsets is obtained.

For a more rigorous and theoretical formulation of the above selection process, the proposed greedy strategy that balances relevance and redundancy by optimizing the combined objective function $\Delta \leftarrow \mathrm{MI}(X,Y) - \gamma \cdot \sum_{i<j}\mathrm{PCC}$.

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TABLE I
NUMER OF CATEGORIES, CORRESPONDING SAMPLES AND TOTAL SAMPLES OF THREE DATASETS

No	Indian Pines		Pavia University		Houston 2013	
	Class name	Samples	Class name	Samples	Class name	Samples
1	Alfalfa	46	Asphalt	6631	Healthy grass	1251
2	Corn-notill	1428	Meadows	18649	Stressed grass	1254
3	Corn-mintill	830	Gravel	2099	Synthetic grass	697
4	Corn	237	Trees	3064	Trees	1244
5	Grass-pasture	483	Painted metal sheets	1345	Soil	1242
6	Grass-trees	730	Bare Soil	5029	Water	325
7	Grass-pasture-mowed	28	Bitumen	1330	Residential	1268
8	Hay-windrowed	478	Self-Blocking Bricks	3682	Commercial	1244
9	Oats	20	Shadows	947	Road	1252
10	Soybean-notill	972			Highway	1227
11	Soybean-mintill	2455			Railway	1235
12	Soybean-clean	593			Parking Lot 1	1233
13	Wheat	205			Parking Lot 2	469
14	Woods	1265			Tennis Court	428
15	Buildings-Grass-Trees-Drives	386			Running Track	660
16	Stone-Steel-Towers	93				
	Total	10249	Total	42776	Total	15029

In Algorithm 2, this strategy iteratively selects the feature subset that maximizes the increment of the objective function, mathematically proving the theoretical optimality boundary of the approximate algorithm.

IV. EXPERIMENTS AND RESULTS ANALYSIS

In this section, the proposed GSDFC method will be compared with eight other band selection methods on three publicly available datasets. At the same time, three publicly available datasets and experimental settings are introduced for the experiments.

A. Datasets

The experiments are conducted on three benchmark public datasets: Indian Pines, Pavia University, and Houston 2013. Table I presents the total number of classes, the number of corresponding samples in each class, and the total number of samples for the three datasets.

1) *Indian Pines:* The Indian Pines dataset is a classic agricultural hyperspectral remote sensing dataset acquired by NASA's AVIRIS sensor in 1992. It covers a region in northern Indiana, USA, with an image size of 145 × 145 pixels and 200 spectral bands ranging from 0.43 to 2.5 μm [54].
2) *Pavia University*: The Pavia University dataset reflects the urban environment of Pavia, Italy, and was acquired by the ROSIS sensor in 2004. This dataset has 103 bands (wavelength range approximately 0.43–0.88 μm) and an image size of 610 × 339 pixels with a spatial resolution of 1.3 m [55].
3) *Houston 2013:* Covering a variety of land cover types, the Houston 2013 dataset includes an HSI with 144 spectral bands ranging from 0.38 to 1.05 μm, while LiDAR data represented by a single band. Both the HSI and LiDAR datasets shared a size of 349 × 1905 pixels with a spatial resolution of 2.5 m. The dataset contained 15 different classes with a total of 15 029 labeled samples [56].

Algorithm 2 Greedy Band Selection based PCC-MI

Require:
$\mathbf{B} = \{1, \ldots, m\}$ {All m bands}
$\mathbf{MI} = [I(X_1; Y), \ldots, I(X_m; Y)]$ {Mutual information between bands and labels}
$\mathbf{PCC} \in \mathbb{R}^{m \times n}$ {Pearson correlation matrix among bands}
k {Number of bands to select}
γ {Redundancy penalty factor}
Ensure:
$S \subseteq \mathbf{B}, |S| = k$ {Selected band set}
$S \leftarrow \emptyset$ {Initialize selected bands}
$C \leftarrow \mathbf{B}$ {Candidate bands}
while $|S| < k$ **and** $C \neq \emptyset$ **do**
  $\Delta_{\max} \leftarrow -\infty$
  $b^* \leftarrow$ null
  for $b \in C$ **do**
    if $S = \emptyset$ **then**
      $\Delta_b \leftarrow \mathbf{MI}[b]$ {Select first band}
    else
      Redundancy $\leftarrow \gamma \cdot \frac{1}{|S|} \sum_{s \in S} |\mathbf{PCC}[b, s]|$
      $\Delta_b \leftarrow \mathbf{MI}[b] -$ Redundancy
    end if
    if $\Delta_b > \Delta_{\max}$ **then**
      $\Delta_{\max} \leftarrow \Delta_b$
      $b^* \leftarrow b$
    end if
  end for
  if $b^* \neq$ null **then**
    $S \leftarrow S \cup \{b^*\}$
    $C \leftarrow C \setminus \{b^*\}$ 27: **end if**
end while
return S

B. Experiments Setting and Parameters Analysis

1) Hardware Configuration: The proposed method is implemented in the PyTorch 2.0 framework and experimentally validated using an NVIDIA RTX 4090-D GPU with 24 GB RAM.

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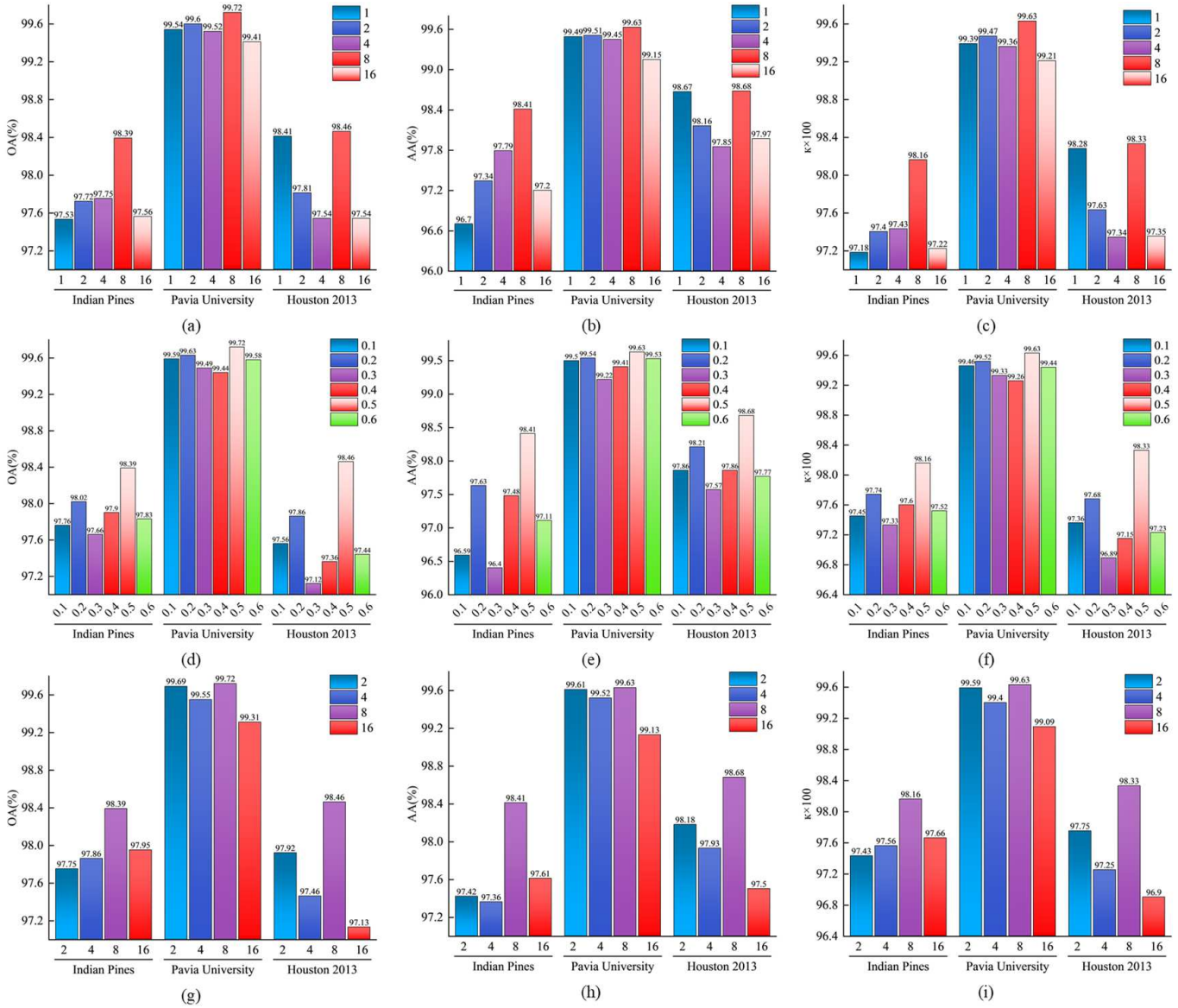


Fig. 5. Classification results under different parameters. (a)–(c) Classification results under different patch sizes. (d)–(f) Classification results under different weight coefficient. (g)–(i) Classification results under different group size.

2) Evaluation Indicator: To better compare the classification performance of different methods, an end-to-end spectral–spatial residual network (SSRN) [57] is employed to classify the selected band subsets of all methods. The SSRN was selected as the benchmark classifier because it is a powerful and widely recognized deep learning model that enables automatic feature extraction, ensuring a fair comparison focused on the quality of the input bands rather than the classifier's capability. In addition, the classification metrics used include OA, AA, and $\kappa \times 100$.

3) Parameters Analysis: The parameters involved in training include training epochs and learning rate set to 10 and $1e^{-3}$, respectively. In the parameter configuration of DFCFFM, the patch size is set to 8, the initial value of the feature fusion weight coefficient is set to 0.5, and the channel group size of GSAM is set to 8. Meanwhile, Daubechies wavelet with order 2 (db2) is utilized to perform DWT. Wavelet decomposition adopts a single-layer decomposition strategy, and the boundary extension method uses periodic extension.

As shown in Fig. 5, the change in the patch size, weight coefficient, and group size have a significant impact on the classification performance of the proposed method. If the patch size and weight coefficient are too large, it will result in the loss of detailed information. Conversely, if the patch size and weight coefficient are too small, it can lead to insufficient global information. When the patch size and the weight coefficient are set to 8 and 0.5, it can achieve the highest classification results, respectively. Meanwhile, different group sizes in GSAM also have a certain impact on classification performance. As the group size increases, computational efficiency and memory consumption will decrease, but at the same time, global feature understanding ability will be affected. As shown in Fig. 5, the best classification performance can be achieved when the group size is 8.

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TABLE II
CLASSIFICATION PERFORMANCE IN ABLATION EXPERIMENTS

Case	Components			Datasets		
	basic	DFCFFM	GBS	Indian Pines	Pavia University	Houston 2013
1	✓			95.25±4.14	98.73±0.49	96.52±1.07
2	✓	✓		98.10±1.47	99.19±0.71	97.47±2.02
3	✓		✓	95.46±3.12	98.99±0.66	96.63±1.65
4	✓	✓	✓	**98.39±1.39**	**99.72±0.23**	**98.46±0.80**

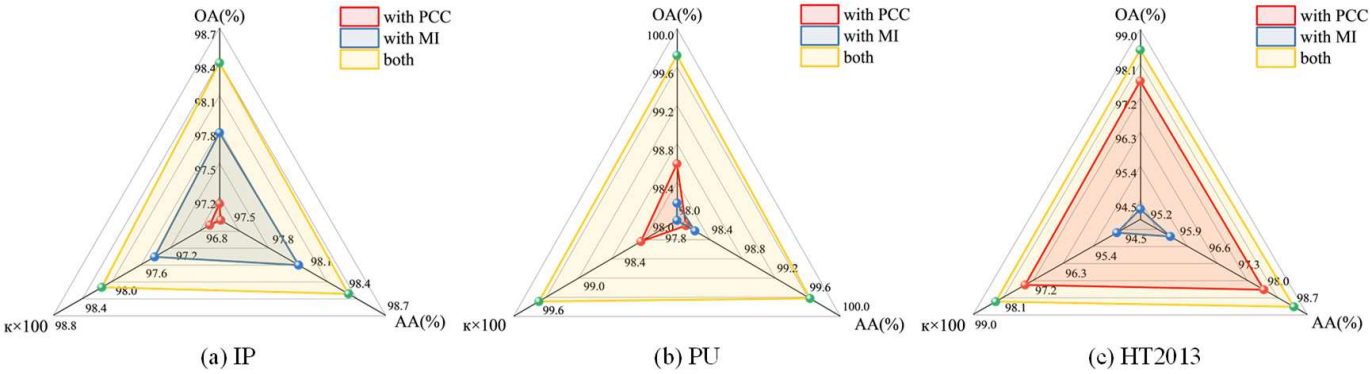


Fig. 6. Classification results obtained by using MI, PCC solely, and combining both. (a) Indian Pines. (b) Pavia University. (c) Houston 2013.

C. Ablation Experiments

To comprehensively evaluate the rationality of combining MI and PCC, and the effectiveness of DFCFFM and GBS in the proposed GSDFC method, some ablation experiments are conducted in this section.

1) Ablation Experiments of DFCFFM: The proposed DFCFFM utilizes DWT to achieved effective fusion of spatial–spectral features in different frequency domains. As shown in case 2 in Table II, the introduction of DFCFFM brings some improvements compared with Case 1. Specifically, the OA values on the Indian Pines, Pavia University, and Houston 2013 datasets increased by 2.85%, 0.46%, and 0.95%, respectively. As shown in case 4, when DFCFFM is combined with GBS, compared with case 1, there is a certain improvement in OA values on the Indian Pines, Pavia University, and Houston 2013 datasets, with increases of 0.29%, 0.53%, and 0.99%, respectively. The experimental results fully demonstrate the effectiveness of DFCFFM.

2) Ablation Experiments of GBS: For band selection, MI can be adopted to evaluate the information contribution of different frequency bands to the target variable, while PCC can be utilized to quickly assess the linear correlation degree between the frequency band and the target variable. The proposed GBS calculates the difference between MI and PCC as the gain and establishes a greedy strategy to select the optimal band subsets. As shown in case 3 in Table II, the introduction of GBS brings some improvements compared with case 1. Specifically, the OA values on the Indian Pines, Pavia University, and Houston 2013 datasets increased by 0.21%, 0.26%, and 0.11%, respectively. As shown in Fig. 6, the rationality and effectiveness of the proposed method have been fully verified.

D. Performance Analysis

In this experiment, nine state-of-the-art band selection methods were selected for comparison with the proposed method, including BSNet-Conv [42], DARecNet-BS [45], EGCSR [43], RFGEC [44], TGSR [48], SPCA_AMGL [58], DropoutCAE [59], SSANet [50], and MOBS-TD [60].

1) Quantitative Assessment via Different Methods: Tables III–V show the OA, AA, $\kappa \times 100$, and accuracy of each class obtained by the proposed method and all comparison methods on three datasets with different numbers of selected bands. The number of selected bands for the Indian Pines, Pavia University, and Houston 2013 datasets are 25, 15, and 20, respectively. For all experiments, 5% of the labeled samples were randomly selected for training, and the rest were used for testing. To ensure statistical significance and mitigate the randomness of the split, this process was repeated ten times with different random partitions. The results demonstrate that the proposed method achieved the highest classification accuracy on all three datasets.

As shown in Table III, the best results for each category in terms of OA, AA, and $\kappa \times 100$ are emphasized in bold. For the Indian Pines dataset, the GSDFC outperforms other methods in single category accuracy. Compared with other methods, the proposed method achieved the highest improvement of 1.38% in OA, 1.88% in AA, and 1.57% in $\kappa \times 100$. As shown in Table IV, TGSR, DropoutCAE, and MOBS-TD achieved relatively high classification performance on the Pavia University dataset. However, compared with other methods, the proposed method outperforms them in all aspects. Compared to MOBS-TD, which achieved the best performance among the comparison methods, the proposed method achieved at least an improvement of 0.19% in OA, an improvement of 0.33% in AA, and an improvement of 0.25% in $\kappa \times 100$. As

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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS USING SELECTED 25 BANDS ON THE INDIAN PINES DATASET WITH 5% TRAINING SIZE

No	BSNet-Conv	DARecNet-BS	EGCSR	RFGEC	TGSR	DropoutCAE	SPCA-AMGL	SSANet	MOBS-TD	GSDFC
1	99.52±1.43	86.93±13.85	92.55±13.14	95.78±8.06	95.98±4.42	92.49±14.93	96.51±6.98	95.30±6.10	89.50±11.90	**99.76±0.71**
2	96.63±3.21	94.93±6.23	96.48±3.83	95.19±4.77	97.58±2.54	93.65±10.84	96.50±3.98	97.37±2.78	96.04±4.58	**97.78±2.64**
3	98.02±1.42	98.37±1.94	**98.96±0.88**	96.55±3.08	97.03±2.82	98.35±1.85	97.46±2.12	98.06±1.54	96.92±3.50	98.71±1.10
4	96.49±3.35	95.59±4.92	96.45±3.48	96.63±1.55	91.93±1.27	93.98±4.59	93.63±5.49	96.48±2.78	96.36±4.17	**98.18±1.36**
5	99.04±1.11	**99.60±0.74**	99.06±1.18	98.74±1.97	99.58±0.73	99.59±0.72	99.58±0.42	99.21±0.78	99.38±0.85	**99.60±0.46**
6	99.28±0.80	99.76±0.53	99.08±1.29	99.20±1.25	99.05±1.06	99.61±0.39	99.47±0.54	99.44±0.98	**99.80±0.30**	99.38±0.62
7	85.68±18.73	**99.60±1.2**	97.37±4.61	98.89±3.33	88.36±19.04	95.38±6.30	95.61±7.15	94.01±9.09	96.65±5.78	97.33±6.75
8	99.34±0.92	99.39±1.25	99.17±1.62	99.86±0.34	99.59±0.71	99.32±1.80	99.39±1.82	98.85±2.88	99.64±1.06	**100**
9	93.51±8.96	**100**	99.41±1.76	93.98±10.59	94.21±8.81	98.42±4.74	89.99±11.96	**100**	93.53±7.01	98.23±2.71
10	94.72±5.07	97.27±1.83	97.69±2.05	96.34±3.94	97.13±3.37	**97.41±1.89**	94.01±5.89	96.98±1.87	96.47±4.49	97.10±2.50
11	98.57±1.01	98.32±1.78	98.52±1.46	98.66±1.00	98.35±1.54	97.41±2.44	97.52±2.35	98.39±1.29	96.89±2.97	**98.85±1.23**
12	96.68±2.75	94.53±3.39	95.19±4.72	96.71±2.87	98.09±1.96	**98.64±1.08**	96.56±4.26	95.86±1.99	97.52±1.74	97.05±2.11
13	99.63±0.66	**100**	99.31±1.19	99.78±0.36	**100**	**100**	99.84±0.34	99.94±0.16	99.64±0.92	99.95±0.16
14	98.53±1.88	98.39±2.14	96.79±3.69	97.97±2.51	98.39±2.01	98.33±2.39	**98.65±1.91**	98.37±2.30	98.04±2.12	98.48±1.99
15	95.37±2.76	96.61±2.89	96.32±3.09	96.94±4.16	**97.24±1.79**	95.33±3.18	94.83±2.73	95.97±2.30	96.17±1.51	96.42±2.07
16	96.52±2.58	96.67±2.42	93.98±3.11	97.02±3.36	95.81±2.82	94.17±3.55	94.91±3.98	96.48±3.02	96.42±2.45	**97.76±1.20**
OA	97.58±1.77	97.48±1.93	97.64±1.87	97.48±2.12	97.82±1.73	97.01±3.45	97.19±2.52	97.91±1.58	97.26±2.28	**98.39±1.39**
AA	96.72±2.39	97.25±1.94	97.27±2.20	97.39±2.07	96.77±2.68	97.01±2.75	96.53±2.85	97.54±1.83	96.81±1.98	**98.41±1.17**
κ×100	97.25±2.01	97.13±2.20	97.30±2.13	97.14±2.42	97.51±1.97	96.59±3.94	96.80±2.87	97.62±1.80	96.87±2.59	**98.16±1.59**

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS USING SELECTED 15 BANDS ON THE PAVIA UNIVERSITY DATASET WITH 5% TRAINING SIZE

No	BSNet-Conv	DARecNet-BS	EGCSR	RFGEC	TGSR	DropoutCAE	SPCA-AMGL	SSANet	MOBS-TD	GSDFC
1	99.27±0.65	99.53±0.40	99.76±0.30	99.48±1.31	99.55±1.02	**99.77±0.26**	99.39±0.93	99.58±0.74	**99.77±0.43**	99.71±0.60
2	99.27±0.50	99.53±0.51	99.72±0.24	99.83±0.19	99.72±0.43	99.70±0.41	99.57±0.61	99.49±0.68	99.82±0.21	**99.94±0.05**
3	**99.68±0.28**	99.00±1.36	96.87±6.79	97.25±5.32	97.10±6.56	97.80±4.43	97.54±3.64	99.08±1.09	99.52±0.40	99.54±0.49
4	99.97±0.06	99.92±0.10	99.70±0.79	99.95±0.06	99.92±0.07	99.97±0.04	99.89±0.11	**99.98±0.03**	99.95±0.08	**99.98±0.03**
5	99.98±0.03	99.97±0..05	99.97±0.10	**100**	99.99±0.02	99.99±0.02	**100**	99.97±0.08	**100**	**100**
6	99.04±1.21	99.18±0.98	99.67±0.50	99.85±0.16	99.88±0.16	99.85±0.17	99.71±0.33	**99.94±0.10**	99.87±0.19	99.84±0.14
7	**99.86±0.34**	98.91±1.21	96.12±4.38	92.18±17.97	99.92±0.25	99.70±0.40	96.94±3.73	99.20±1.47	96.42±10.46	99.26±0.71
8	97.65±1.49	97.73±2.01	98.57±0.84	98.67±0.84	**98.69±1.20**	97.81±1.22	98.24±0.95	98.05±1.92	98.35±1.18	98.39±0.98
9	**100**	99.99±0.03	**100**	99.98±0.07	99.92±0.17	**100**	99.97±0.05	**100**	**100**	**100**
OA	99.22±0.41	99.33±0.49	99.34±0.58	99.03±1.92	99.49±0.49	99.49±0.33	99.29±0.52	99.45±0.52	99.53±0.53	**99.72±0.23**
AA	99.41±0.29	99.31±0.46	98.93±0.90	98.55±2.69	99.41±0.69	99.40±0.52	99.03±0.60	99.48±0.42	99.30±1.20	**99.63±0.27**
κ×100	98.97±0.54	99.11±0.66	99.12±0.76	98.72±2.52	99.33±0.64	99.33±0.44	99.05±0.69	99.27±0.70	99.38±0.70	**99.63±0.31**

shown in Table V, among all methods, the proposed method achieved the best classification performance in terms of OA, AA, and $\kappa \times 100$ on the Houston 2013 dataset. In particular, the proposed method achieved the best classification accuracy in single category accuracy.

2) Visual Classification Results Analysis: Figs. 7–9 present the original ground truth and the visualized classification results of all methods on the three datasets. It is evident that the proposed method produces visual classification results that are closer to the original ground truth on all three datasets, demonstrating better classification performance. On the Indian Pines dataset (Fig. 7), most methods misclassify Alfalfa and Grass pasture. While the proposed method also makes some misclassifications, it exhibits the fewest misclassified regions compared to other methods. On the Pavia University dataset (Fig. 8), BSNet-Conv and TGSR misclassify a large area of the Gravel class. Compared with other methods, the proposed method maintains higher regional consistency and provides a visualization that is closer to the ground truth. Unlike the previous two datasets, on the Houston 2013 dataset (Fig. 9), most methods achieved high regional consistency. Compared with other methods, the proposed method not only maintains high regional consistency but also achieves better edge information localization. Overall, the proposed method demonstrates strong classification performance on all three datasets. Both quantitative analysis and visual results have fully verified the effectiveness of the proposed method.

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS USING SELECTED 20 BANDS ON THE HOUSTON 2013 DATASET WITH 5% TRAINING SIZE

No	BSNet-Conv	DARecNet-BS	EGCSR	RFGEC	TGSR	DropoutCAE	SPCA-AMGL	SSANet	MOBS-TD	GSDFC
1	96.72±2.64	95.53±3.89	97.10±2.73	95.35±3.24	97.62±2.44	98.79±1.22	93.96±3.27	94.60±6.66	98.58±1.74	**99.60±0.63**
2	99.25±0.90	98.70±2.37	99.45±1.04	97.67±3.43	99.81±0.22	99.58±0.50	99.50±0.64	97.08±3.14	99.69±0.24	**99.91±0.16**
3	99.98±0.04	99.98±0.05	**100**	**100**	99.95±0.14	**100**	96.84±6.41	99.98±0.05	99.64±1.07	98.96±0.43
4	98.88±0.80	98.89±1.28	97.86±1.63	97.76±2.62	97.63±3.54	99.06±0.88	97.91±2.66	97.85±3.78	**99.49±0.55**	98.76±0.90
5	99.80±0.37	99.31±0.99	99.36±0.92	98.82±1.25	99.64±0.62	**99.86±0.03**	99.37±0.71	98.48±1.30	99.71±0.30	99.82±0.35
6	99.83±0.41	98.67±1.75	99.52±0.61	99.70±0.91	**99.97±0.10**	99.03±1.94	99.59±0.42	**99.97±0.10**	99.25±0.45	99.76±0.38
7	95.99±3.46	96.67±3.10	96.26±3.03	95.76±2.30	97.11±2.15	97.75±1.58	96.56±1.92	94.45±3.02	95.84±3.00	**98.36±1.22**
8	97.12±2.87	96.13±5.63	98.75±2.13	98.46±1.96	98.88±2.37	98.34±1.70	97.36±3.60	97.73±2.06	98.12±4.01	**98.92±0.85**
9	95.35±2.96	96.10±4.09	93.16±5.51	92.77±5.15	97.02±3.19	96.83±1.86	93.55±5.13	93.67±3.17	96.94±4.88	**97.06±1.81**
10	91.83±7.04	94.20±4.43	89.92±7.14	90.33±7.17	93.16±3.22	95.79±3.31	92.56±3.61	91.60±7.45	92.87±4.19	**96.75±2.16**
11	95.15±5.96	93.73±7.02	96.92±2.61	91.83±6.02	96.25±6.05	97.26±1.71	94.46±6.36	93.57±4.91	96.15±4.13	**98.29±0.78**
12	93.45±5.81	90.91±5.57	92.04±8.72	92.46±7.43	94.87±4.77	94.18±5.11	94.04±4.26	88.98±9.04	95.07±3.61	**95.33±4.07**
13	95.59±11.17	97.77±1.45	98.57±1.46	**99.39±1.11**	98.52±0.96	98.94±1.41	98.02±2.09	98.24±1.36	98.51±3.81	98.77±1.14
14	99.41±1.39	97.99±2.11	99.45±1.39	98.04±2.39	99.16±1.60	99.38±1.47	98.79±1.20	97.14±2.67	99.35±1.30	**99.97±0.08**
15	99.24±0.97	99.92±0.17	99.85±0.45	99.82±0.44	99.83±0.40	99.90±0.21	99.97±0.07	99.31±1.21	**100**	99.98±0.05
OA	96.58±2.39	96.38±2.32	96.52±2.53	95.73±2.52	97.50±1.72	97.80±1.09	96.23±2.36	95.22±2.82	97.52±1.42	**98.46±0.80**
AA	97.17±2.08	96.97±1.85	97.21±1.85	96.54±2.00	97.96±1.40	98.31±0.96	96.83±2.01	96.18±2.06	97.95±1.23	**98.68±0.69**
κ×100	96.31±2.58	96.08±2.51	96.24±2.74	95.38±2.72	97.30±1.86	97.84±1.18	95.92±2.55	94.84±3.05	97.31±1.54	**98.33±0.87**

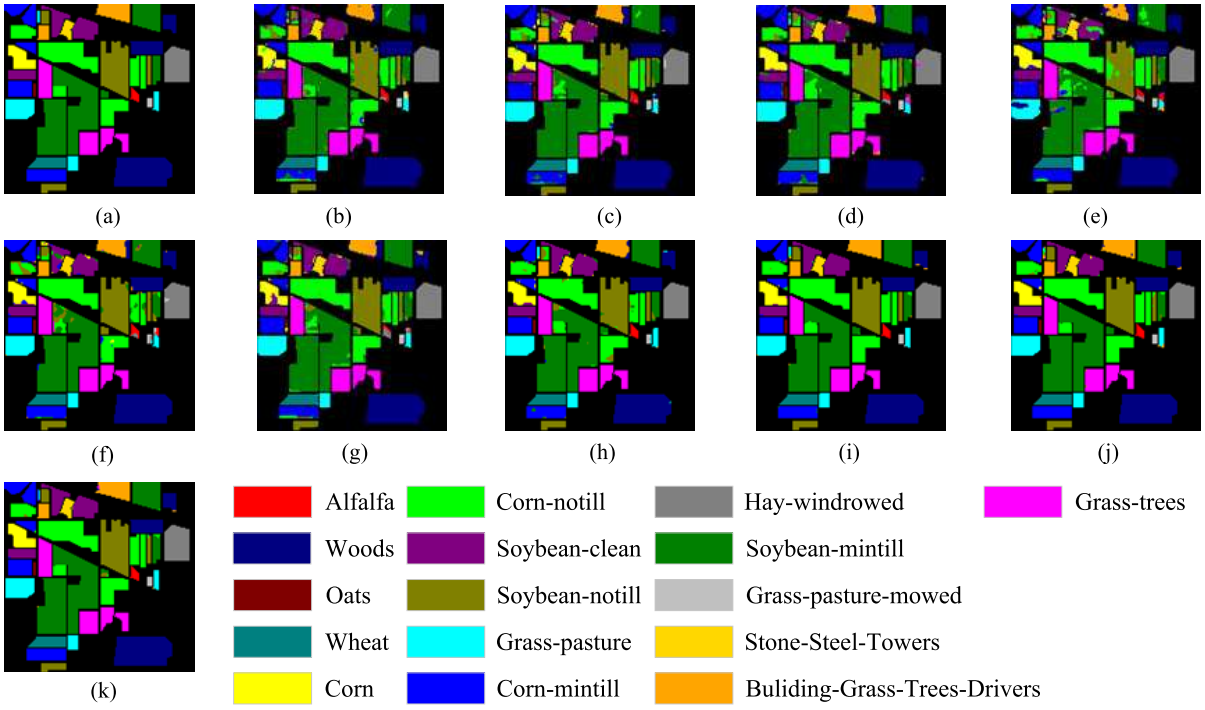


Fig. 7. Visual classification results of different methods on Indian Pines dataset. (a) gt. (b) BSNet-conv. (c) DARecNet-BS. (d) EGCSR. (e) RFGEC. (f) TGSR. (g) DropoutCAE. (h) SPCA-AMGL. (i) SSANet. (j) MOBS-TD. (k) GSDFC.

E. Quantitative Analysis of Band Selection Number

By varying the number of selected bands, this experiment aims to compare the classification performance of all methods. To thoroughly validate the effectiveness of the selected bands, we compare the classification performance against using all original bands under the identical SSRN framework. Specifically, the range of selected bands is [10:50]. For each dataset, two sets of comparative results are presented: 1) the OA classification results of all methods under different numbers of selected bands and 2) on three datasets, the average OA, AA, and $\kappa \times 100$ of different band selection methods within the selected band range.

1) OA Classification Results of All Methods Under Different Numbers of Selected Bands: As shown in Fig. 10, the dashed line represents the classification accuracy under selected all bands, and the solid line represents the OA classification

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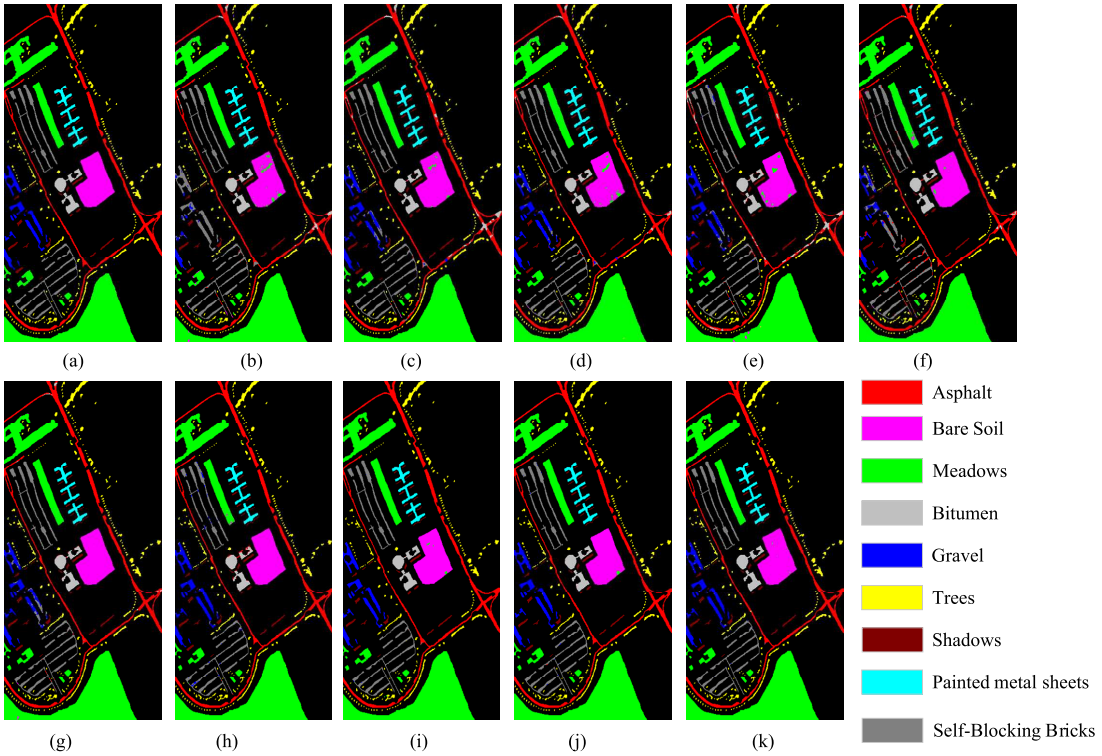


Fig. 8. Visual classification results of different methods on Pavia University dataset. (a) gt. (b) BSNet-conv. (c) DARecNet-BS. (d) EGCSR. (e) RFGEC. (f) TGSR. (g) DropoutCAE. (h) SPCA-AMGL. (i) SSANet. (j) MOBS-TD. (k) GSDFC.

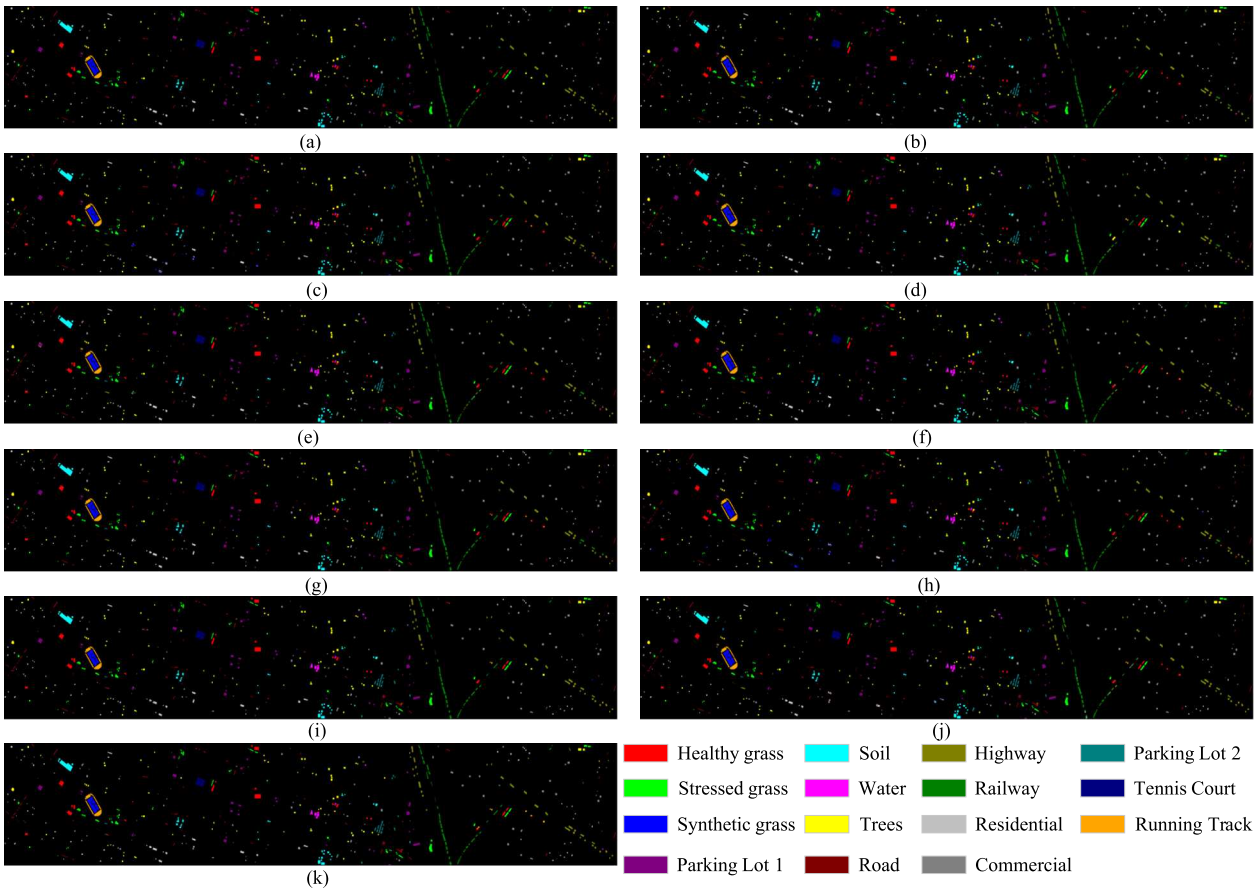


Fig. 9. Visual classification results of different methods on Houston 2013 dataset. (a) gt. (b) BSNet-conv. (c) DARecNet-BS. (d) EGCSR. (e) RFGEC. (f) TGSR. (g) DropoutCAE. (h) SPCA-AMGL. (i) SSANet. (j) MOBS-TD. (k) GSDFC.

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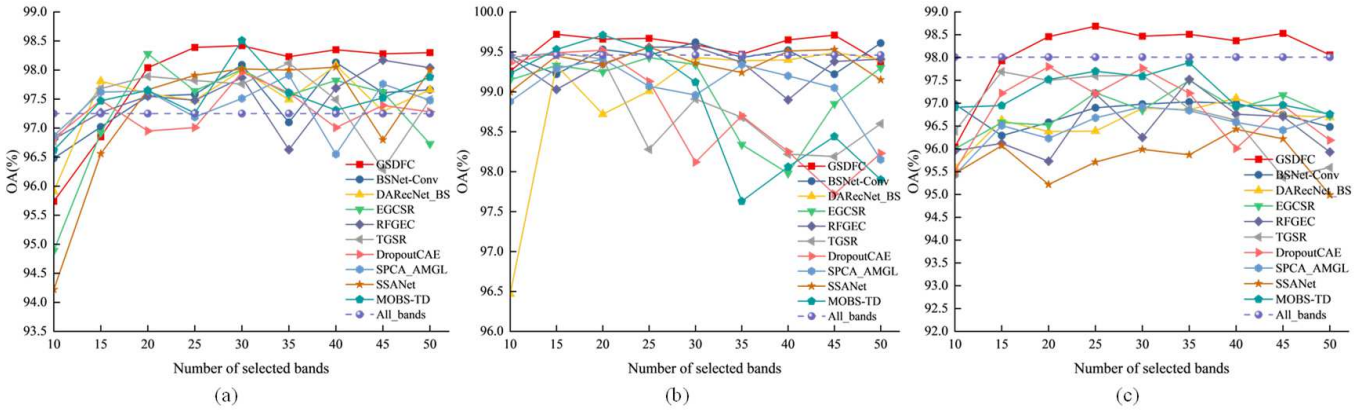


Fig. 10. Comparison of OA under different number of bands on (a) Indian Pines, (b) Pavia University, and (c) Houston 2013, respectively.

TABLE VI
AVERAGE CLASSIFICATION PERFORMANCE WITHIN THE SPECIFIED SELECTED BAND RANGE

Datasets	indices	GSDFC	BSNet-Conv	DARecNet-BS	EGCSR	RFGEC	TGSR	Dropout CAE	SPCA_AMGL	SSANet	MOBS-TD	All bands
Indian Pines	OA	**97.96**	97.47	97.37	97.28	97.50	97.48	97.28	97.39	97.25	97.43	97.25
	AA	**97.77**	97.26	97.07	96.81	97.34	97.06	96.91	96.90	96.91	97.25	95.31
	κ×100	**97.54**	97.11	97.01	96.89	97.11	97.01	96.90	96.92	96.75	97.08	96.84
Pavia University	OA	**99.57**	99.45	98.96	99.00	99.33	98.80	98.72	99.04	99.35	98.79	99.46
	AA	**99.43**	99.40	99.07	99.09	99.06	99.04	98.85	98.97	99.25	99.11	99.31
	κ×100	**99.42**	99.28	98.61	98.67	99.12	98.40	98.30	98.72	99.14	98.39	99.29
Houston 2013	OA	**98.12**	96.78	96.59	96.83	96.47	96.81	96.88	96.48	95.77	97.25	98.01
	AA	**98.38**	97.33	97.13	97.36	97.08	97.44	97.46	97.06	96.48	97.72	98.37
	κ×100	**97.94**	96.51	96.31	96.57	96.18	96.55	96.61	96.20	95.43	97.02	97.84

accuracy of all methods under different numbers of selected bands. On the one hand, the proposed method achieved the highest OA on all three datasets. On the other hand, when a specific number of bands is selected, as the number of selected bands increases, compared with other methods, the classification performances are relatively stable, which proves the robustness of the proposed method.

2) Average Classification Performance of Different Methods Within the Selected Band Range: The same conclusion as Fig. 10 can be drawn from Table VI, that is, the proposed method outperforms other comparative methods in terms of average OA, AA, and $\kappa \times 100$. Specifically, on the three datasets, the proposed method achieved at least 0.46%, 0.12%, and 0.87% OA improvements, respectively. Meanwhile, when the number of selected bands for the three datasets are 25, 15, and 20, the resulting classifications exceeded the average classification accuracy and the classification accuracy under all bands.

Thus, the number of selected bands are determined to set as 25, 15, and 20 for the Indian Pines, Pavia University, and Houston 2013 datasets, respectively.

F. Selected Bands Analysis

In this section, the selected bands will be analyzed based on their positions and entropy values. The indices of the optimal bands selected by various methods for the three datasets are summarized in Table VII. Figs. 11–13 present the distribution maps of the selected bands, as well as the entropy maps corresponding to each band.

As shown in Figs. 11–13, the upper part presents the spectral locations of the bands specified in Table VII. Each row corresponds to a specific band selection method and the spectral locations of the selected band. The lower part displays the entropy values of the entire spectral band. It can be observed that on the Indian Pines dataset (Fig. 11), RFGEC predominantly selects bands within the [15:30] and [35:45] intervals, while neglecting the bands with higher entropy values in the [145:200] intervals. Although the bands selected by TGSR are relatively dispersed, it can be observed that the selected bands within a certain interval have lower entropy values. Similarly, in the interval of [75:105], BSNet-Conv and DARecNet-BS commonly selected a significant number of bands with lower entropy values. The SPCA_AMGL method predominantly selects bands within the [35:200] interval, while neglecting bands with higher entropy values in the [0:30] interval. On the Pavia University dataset (Fig. 12), although the selected bands by BSNet-Conv, DARecNet-BS, EGCSR, TGSR, and DropoutCAE are relatively dispersed, they all selected bands with lower entropy values in the [65:75] interval, which contributed to a decrease in classification performance. The selected bands of SPCA_AMGL have high entropy values and

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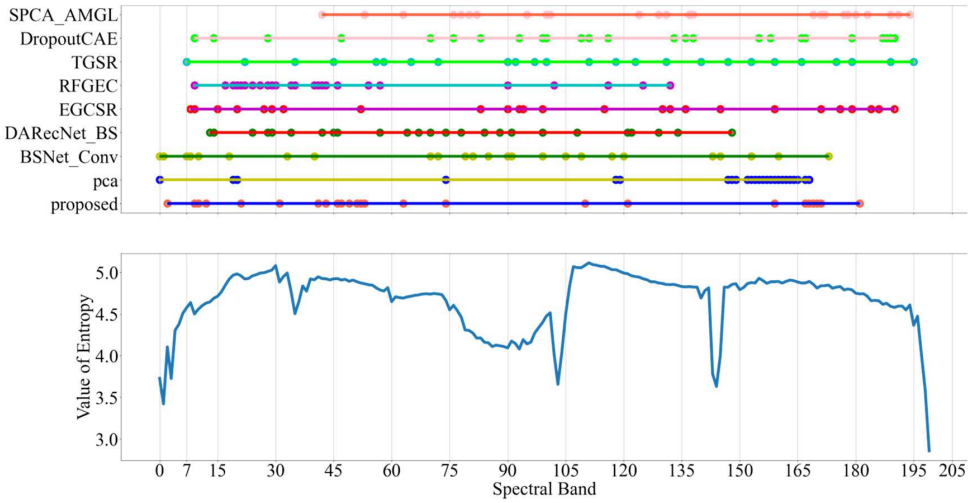


Fig. 11. Best 25 bands of Indian Pines dataset selected by different BS methods (above) and the entropy value of each band (below).

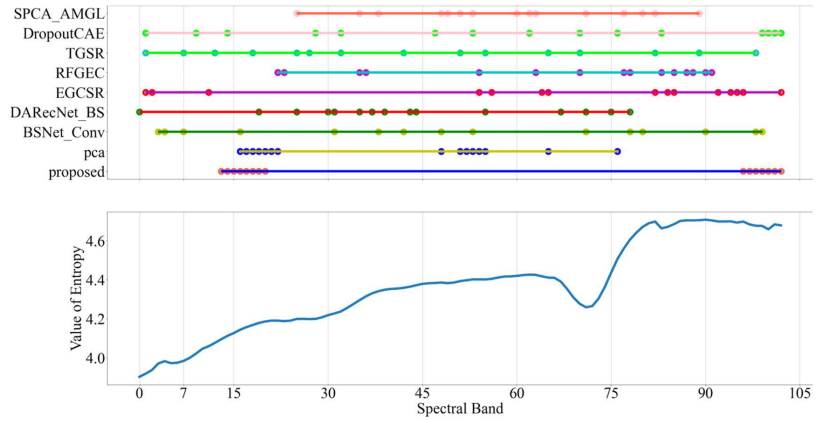


Fig. 12. Best 15 bands of Pavia University dataset selected by different BS methods (above) and the entropy value of each band (below).

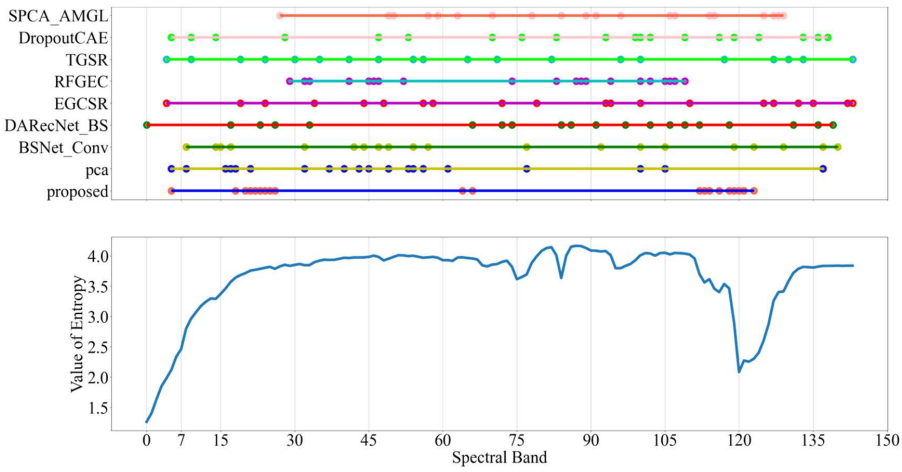


Fig. 13. Best 20 bands of Houston 2013 dataset selected by different BS methods (above) and the entropy value of each band (below).

are evenly distributed; however, it neglects a large number of bands with high entropy values in the [90:103] interval. On the Houston 2013 dataset (Fig. 13), the bands selected by RFGEC are concentrated in the [25:110] interval, neglecting bands with higher entropy values in the [130:144] interval. The bands selected by SSANet are concentrated in the [37:54], with a dense distribution, which can lead to significant redundancy.

On the one hand, due to the high correlation between adjacent bands, selecting denser bands will lead to greater information redundancy. On the other hand, bands with

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TABLE VII
SELECTED BANDS BY TEN METHODS ON THREE DATASETS

Datasets	Methods	Selected Band numbers
Indian Pines	GSDFC	[21,52,181,74,169,46,171,12,43,121,31,47,159,41,110,2,51,170,63,10,168,49,167,53,9]
	BSNet-Conv	[80,97,43,71,72,7,92,151,134,87,100,15,10,84,95,38,106,122,3,34,37,48,36,86,190]
	DARecNet_BS	[58,67,157,61,70,89,71,18,35,34,14,172,97,25,77,74,20,86,146,48,21,3,5,109,66]
	EGCSR	[94,115,20,132,9,83,15,190,184,52,90,99,171,32,159,93,8,145,130,179,186,29,176,136,27]
	RFGEC	[30,132,41,40,102,9,34,20,57,35,46,21,29,116,54,43,24,125,26,42,28,90,19,17,22]
	TGSR	[7,22,35,45,56,58,65,72,90,92,97,100,111,118,123,131,140,147,153,159,166,175,179,189,195]
	DropoutCAE	[9,14,28,47,70,76,83,93,99,100,109,111,116,133,136,138,155,158,166,67,179,187,188,189,190]
	SPCA_AMGL	[191,180,78,124,138,171,169,189,131,82,101,63,177,76,183,95,129,42,172,100,137,178,194,80,53]
	SSANet	[65,33,67,64,32,73,6,18,25,68,122,179,150,19,155,95,119,197,63,47,71,82,34,0,66]
	MOBS-TD	[7,14,19,25,30,31,38,42,47,49,56,61,71,74,86,98,114,125,128,136,150,166,171,182,189]
Pavia University	GSDFC	[73, 1, 98, 8, 88, 58, 97, 6, 100, 7, 83, 64, 11, 92, 2]
	BSNet-Conv	[90,42,16,48,71,3,78,38,80,53,7,31,4,99,98]
	DARecNet_BS	[30,75,44,55,25,19,39,0,78,37,31,35,67,71,43]
	EGCSR	[54,65,85,64,84,96,1,2,56,94,11,82,92,95,102]
	RFGEC	[63,23,78,35,77,83,91,70,90,88,85,54,87,36,22]
	TGSR	[1,7,12,18,25,27,32,42,51,55,65,70,82,89,98]
	DropoutCAE	[1,9,14,28,32,47,53,62,70,76,83,99,100,101,102]
	SPCA_AMGL	[63,49,82,77,38,25,35,53,89,60,51,48,80,62,71]
	SSANet	[51,73,63,32,71,25,52,6,70,1,47,82,20,9,94]
	MOBS-TD	[5,7,15,19,34,35,56,65,69,73,83,84,89,99,102]
Houston 2013	GSDFC	[21,121,66,118,5,20,114,24,123,22,116,23,120,18,112,25,64,119,26,113]
	BSNet-Conv	[54,15,44,105,8,47,77,17,123,100,137,57,140,92,119,49,129,42,14,32]
	DARecNet_BS	[102,86,74,17,0,84,109,106,131,26,136,23,118,97,72,112,66,33,139,91]
	EGCSR	[48,72,58,100,19,44,93,24,132,143,56,94,127,79,110,34,135,142,125,4]
	RFGEC	[45,74,88,106,46,89,52,87,100,47,29,94,83,107,32,33,102,41,109,105]
	TGSR	[4,9,19,24,30,35,41,47,54,56,65,71,82,96,100,117,127,130,133,143]
	DropoutCAE	[5,9,14,28,47,53,70,76,83,93,99,100,102,109,116,119,124,133,136,138]
	SPCA_AMGL	[107,91,84,57,70,89,114,63,127,125,27,50,49,115,96,106,59,129,128,78]
	SSANet	[143,142,51,50,49,48,47,46,45,44,43,42,41,40,39,38,37,52,53,54]
	MOBS-TD	[9,15,33,34,48,56,71,76,84,89,92,99,102,108,109,110,114,117,128,137]

higher entropy values contain more information, and selecting bands with lower entropy values will limit classification performance. The proposed method, which integrates MI and PCC, comprehensively accounts for both linear and nonlinear correlations among bands. Consequently, the selected bands in the first two datasets are relatively dispersed and have higher entropy values. Although the selected bands on the Houston 2013 dataset are denser, most of the selected bands have high entropy values. Therefore, compared with other methods, the bands selected by the proposed achieve significantly better classification performance than other advanced methods.

G. Comparison of Computational Complexity

To comprehensively evaluate the performance of each method, three metrics, running time, Params, and FLOPs, were employed to compare different models. As shown in Table VIII, significant differences in efficiency and computational cost can be observed among the methods. The

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TABLE VIII
COMPARSION OF COMPUTATIONAL COMPLEXITY OF VARIOUS METHODS ON THREE DATASETS

Datasets	Metric	GSFDC	BSNetconv	DARecNet_BS	DropoutCAE	SPCA_AMGL	SSANet
Indian Pines	Time(s)	**9.32**	49.22	120.31	53.57	49.8	39.07
	Params(K)	158.62	590.29	200.98	**29.33**	1650.03	683.86
	FLOPs(M)	12.53	15.13	527.99	**0.03**	3860.5	2529.23
Pavia University	Time(s)	60.26	122.18	272.69	204.28	**43.58**	200.18
	Params(K)	86.68	397.65	163.93	**7.82**	1615.48	335.78
	FLOPs(M)	6.45	10.32	271.27	**0.01**	2000.12	1181.43
Houston 2013	Time(s)	60.91	**53.39**	141.85	97.58	46.69	885.44
	Params(K)	117.24	479.07	176.83	**10.85**	1628.83	429.71
	FLOPs(M)	9.05	12.35	379.64	**0.01**	2792.96	1653.03

computational complexity comparison was conducted among methods implemented in the same programming environment to ensure a fair and controlled comparison. The other advanced methods, such as TGSR, which were implemented in MATLAB, were excluded from this specific complexity analysis.

Specifically, the proposed GSFDC demonstrates highly competitive overall efficiency. Although it does not achieve the absolute shortest running time on all datasets, it consistently offers an excellent balance between running time and model complexity. Specifically, GSFDC obtains the shortest running time on the Indian Pines dataset. More importantly, GSFDC uses significantly fewer parameters and computations than many larger models. While DropoutCAE achieves the lowest Params and FLOPs, its running time are considerably longer than GSFDC. In contrast, methods such as SPCA_AMGL and SSANet demand excessively high memory and computational resources. These results confirm that GSFDC offers a favorable balance between performance and efficiency.

V. CONCLUSION

To better learn spatial–spectral information across different frequency domains, effectively capture both global and local features, and ultimately select the optimal subset of bands among all spectral bands, in this article, a GSDFC method is proposed for unsupervised band selection of HSIs. On the one hand, a DFCFFM is constructed to perform feature learning on the high-frequency and low-frequency components of patch data, and the learned features in different frequency domains are extracted and fused to capture the global and local features of HSIs. On the other hands, by combining the greedy selection strategy with PCC and MI, the optimal subset of bands is selected from all bands, and the classification performance of the selected subset of bands is verified using the SSRN classifier.

Despite the proposed method achieving high overall classification performance on all three datasets, there are still certain limitations. The proposed DFCFFM has a lower computational complexity when dealing with few-shot datasets, but it requires a large amount of memory and has a higher computational complexity when handling large datasets such as WHU_Hi_HanChuan. Therefore, in future work, efforts will focus on reducing the memory and computational costs of large-scale hyperspectral data processing while ensuring performance through some lightweight model design and optimization strategies.

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Zexin Zeng received the bachelor's degree in electronic information engineering from Guangzhou Maritime University, Guangdong, China, in 2021, and the master's degree in communication and information systems from Qiqihar University, Qiqihar, China, in 2024.

His research interests include hyperspectral image processing and machine learning.



Cuiping Shi (Member, IEEE) received the M.S. degree in signal and information processing from Yangzhou University, Yangzhou, China, in 2007, and the Ph.D. degree in information and communication engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2016.

From 2017 to 2020, she held Post-Doctoral Research with the College of Information and Communications Engineering, Harbin Engineering University, Harbin. From 2021 to 2023, she was a Professor with the Department of Communication Engineering, Qiqihar University, Qiqihar, China. since 2024, she has been working with the College of Information Engineering, Huzhou University, Huzhou, China. Her main research interests include remote sensing image processing pattern recognition and machine learning.



Liguo Wang (Member, IEEE) received the M.S. and Ph.D. degrees in signal and information processing from Harbin Institute of Technology, Harbin, China, in 2002 and 2005, respectively.

From 2006 to 2008, he held a post-doctoral research position with the College of Information and Communications Engineering, Harbin Engineering University, Harbin, where he is currently a Professor. Since 2020, he has been working with the College of Information and Communication Engineering, Dalian Nationalities University, Dalian, China. His main research interests include remote sensing image processing and machine learning.



Haizhu Pan received the M.S. degree in control engineering from Qiqihar University, Qiqihar, China, in 2009, and the Ph.D. degree in control engineering from Harbin Engineering University, Harbin, China, in 2018.

She is currently a Professor with the College of Computer and Control Engineering, Qiqihar University. Her research interests includeintelligent computing, machine learning, and hyperspectral image processing.

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